

Gio SO

Automation & Industrial AI Engineer

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SUMMARY

Automation and Industrial AI Engineer (Valedictorian), specialized in Mechatronics, Control Systems, and Industry 4.0/5.0. Successful first industrial experience within the Automation and R&D team at Holweg Weber, contributing to the optimization and safety of production lines (bag-making machines) in close collaboration with electrical, mechanical, instrumentation teams as well as stakeholders. Experienced in managing end-to-end automation sub-projects: requirements analysis, sensor selection and integration, Siemens PLC programming (Simotion, TIA Portal), OPC UA communication setup, HMI/Node-RED development, validation testing in the workshop, and deployment on embedded systems (Edge Gateway).

EDUCATION

UNIVERSITY OF STRASBOURG

Master's degree, Mechatronics, Energy and Intelligent Systems

Strasbourg, France

Sep 2023 - Jul 2025

Valedictorian, 1st / 17

- **Focus:** Industrial automation, embedded systems, project management, C++/Python programming, sensors and instrumentation, multimodal AI models, time-series prediction, anomaly detection, computer vision

UNIVERSITY OF STRASBOURG

Bachelor's degree, Applied Physics and Engineering Science

Strasbourg, France

Sep 2020 - Jul 2023

WORK EXPERIENCE

HOLWEGWEBER GROUP

AI and R&D Automation Engineering Intern

Molsheim, France

Feb 2025 - Jul 2025

- **Data Science Engineering:** Led the end-to-end data lifecycle, including real-time acquisition and processing from industrial sensor data (vibration, acoustic, motor torque) using OPC UA, MQTT, and IO-Link. Performed extensive EDA and data preprocessing.
- **AI Engineering & Model Training:** Designed, trained, and validated diverse Deep Learning models (VAE, CNN, Few-Shot Learning) for high-frequency time-series and spectral signals.
- **Digital Innovation & Deployment:** Engineered a scalable, containerized deployment solution (Docker, Rest Flask API), implementing and testing the model on an Edge Gateway (Belden OpEdge).
- **System Maintenance:** Designed an HMI for both diagnostic reporting and critical on-site model re-training to ensure the AI system's maintainability and immediate adaptation.

UNIVERSITY PROJECTS

PREDICTIVE MAINTENANCE and ANOMALY DETECTION in MILLING PROCESS

Sep 2024

- Implementation of a high-frequency vibration signal acquisition and processing pipeline to monitor cutting tool status and machine health using Python and VSCode.
- Development of an anomaly detection framework to anticipate process drifts and improve machined part quality in real-time.

INTELLIGENT ROBOT NAVIGATION (ROS2)

Févr 2024

- Design of an automation and control architecture for a mobile robot: control chains, sequential logic (state management), local supervision, and alarm handling.
- Integration of sensors (LiDAR, odometry) into a control loop for trajectory planning and obstacle avoidance, with validation tests in near-real conditions.

ACTIVITY

COMPETITIVE SWIMMING

2013 - Present

- Ranked National 3

ADDITIONAL INFORMATION

Technical Skills: Automation & Control (TIA Portal, Simotion), Industrial Communication (OPC UA, Node-RED, MQTT), Python (PyTorch, TensorFlow, SkLearn), C++, SQL, Git/GitHub, Docker.

Languages: French (C2 - Native), English (C2 - Highly Fluent), German (A2), Korean (B1).